absolutely no good guess about the behavior of the function, and equates to wasting information otherwise.

We will see what different options there are when it comes to the mean and covariance functions, and how they impact the resulting GP model. Different from Chapter 2, we will take a hands-on approach here and revolve our discussions around code implementation in Python. By the end of this chapter, we will have developed a pipeline for selecting appropriate mean and covariance functions as well as optimizing their hyperparameters.

3.1 Question: Why can't you seem to change some people's mind? Answer: Because of their priors.

To show how important priors are in a Bayesian model, we will consider the following scenario.

Let's say we are hanging out with our friends, Bob and Alice, at a carnival, and we are talking to someone who claims that she's a psychic. The way she allows us to test this claim is the following procedure: each of us thinks of a number between 0 and 9, and she will tell us what specific number we are thinking of, and we can repeat this for however many times as we'd like.

Now, all three of us are curious about this supposed psychic, and we decide to conduct this test one hundred times. Amazingly, after these one hundred tests, the woman at the carnival guesses the numbers we think of correctly. However, after we are done, we all have different reactions, as seen in Figure 3.1.

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